*Gemini-AI generated logo

BUBBLE LETTERS

A LEARNING GAME DESIGN FOR K~4 READING LITERACY

by Darryl Hutchinson

Problem

US Reading Literacy Rates for K~4

- **~37%** of US students are reading below BASIC level. (NAEP, 2022)
- **~70%** of low-income 4th grade students are below BASIC level. (NLI, 2022-2023)
- Children not proficient in reading by 3rd grade are **4x more likely to drop out** of high school (Casey Foundation, 2011)

Learning Theory

Science of Reading Instruction

- Science of Reading Literacy (AAEI, 1836), (NRP, 2000)
  - Phonological Awareness, Phonics
- Decoding (Crawford, 2024)

Interactive Assessment

- Stealth Assessment (Shute, Kim, 2005)
- Games as Assessment (Gee, 2010)
- Gameful Assessment (Kocadere, 2015)
- Evidence Centered Design (Mislevy, 2004), (Arieli-Attali, 2019)

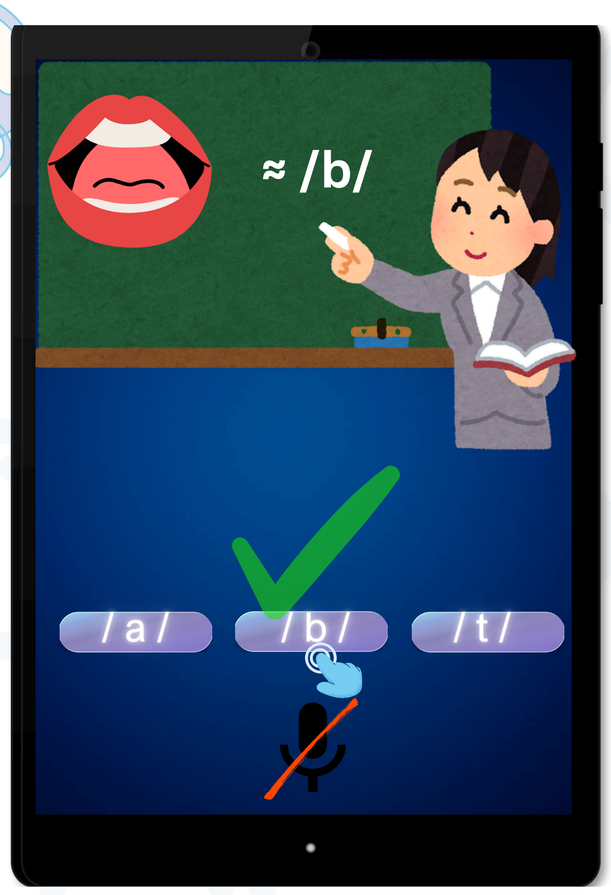
Game Based Learning Affordances

- Situated Feedback (Plass, Mayer, et.al. 2019)
- Self Determination Theory (2000, 2023)
- MDA Framework (2004)
- Multiple Representation Principles (Ainsworth, 2021)

Design Solution:

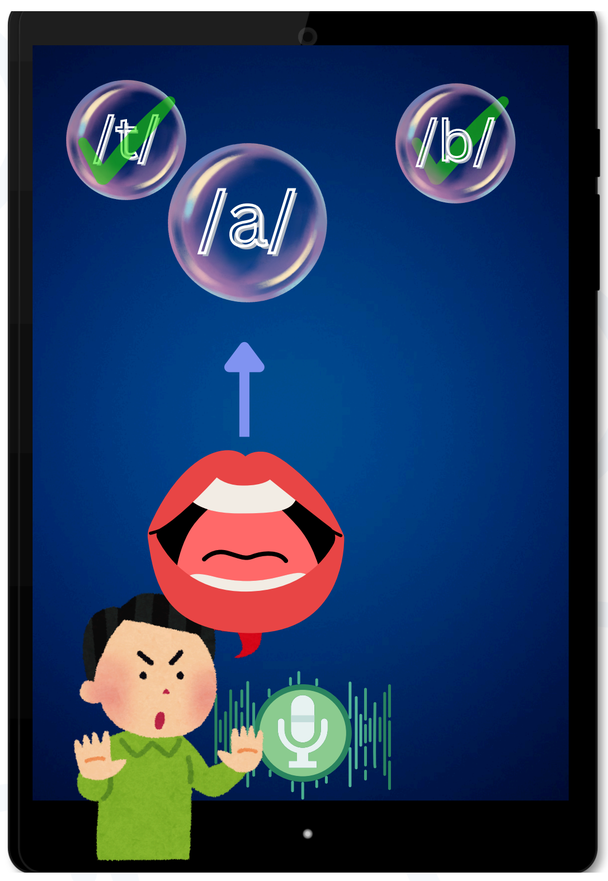
Combining the latest research on the **science of reading**, **interactive assessment**, and **game based learning** affordances, 'Bubble Letters' utilizes **scaffolding**, **voice** recognition and **touch** gestures empowering learners to practice reading skills by making and **playing with phoneme bubbles!**

Phase 1: Listen & Learn



The experience begins with simple **phonetics instruction**, **modeling** proper mouth position and sound for each phoneme.

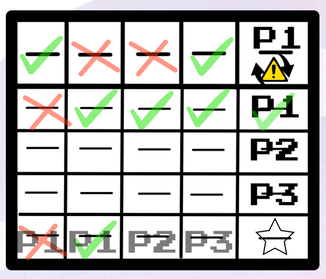
Phase 2: Speak!

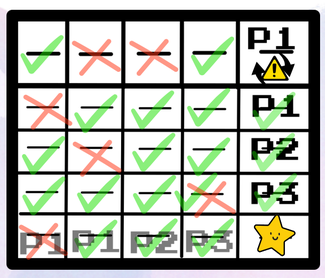


With microphone input enabled, the learner can create **phoneme bubbles** with their own **voice**, **practicing speaking** and building confidence.

Phase 3: Match & Master

With more freedom and challenge the learner can **play with their created phoneme bubbles,** make grapheme matches and demonstrate mastery.





Stealth assessment helps populate a background log of learner success and gracefully determines progress through phases while also providing a reference to support teachers, parents and students.

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G4L | MASTER OF SCIENCE GAMES FOR LEARNING
Where play has serious impact.